

Assessment - Practical Evaluation: Subject

1. [4MCSA](#)
2. Assessment - Practical Evaluation: Subject

Conditions d'achèvement

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À rendre : lundi 1 juin 2026, 23:59



1 - Introduction

This project has to be realized by groups of maximum 2 students in 2 weeks, and is mostly based on what you have seen during this course but some parts may need a little research to complete. **One and only one** group of 3 is allowed **if and only if** the class has an odd number of students and all other groups have 2 members.

2 - Presentation

You are working in a company specialised in Microsoft products. Your new client is *MedSearch*. They are specialised in medical research and are based in Caen and Saint-Cénéri-le-Gerei.

In fact, they do not have a centralized information system. People have their own computers and use USB keys to share documents. Only one employee oversees the information system. You have to pay attention to this last point in your project.

They have many requirements like WSUS, WDS, compute container system, DFS replication, mail servers, collaborative solution, etc. But at this time, they cannot run any of these services because the basic infrastructure does not exist. You need to provide a basic infrastructure that will be able to support all requirements.

3 - Process

3.1 - High availability

The client needs a virtualization infrastructure that will host all future services. As Microsoft Solution Experts, you will suggest a Hyper-V solution. This solution must be highly available at many points:

- Networks:
 - Management
 - Storage
 - VM
- Storage
 - You must find a solution for redundancy in case of disk failure or all the storage solution failure
- Hosts
 - They must be clustered

3.2 - IIS websites

For each new research project, *MedSearch* ask their original provider to install a website specifically designed for them. This website allows people to share research progress and is only used for the duration of the project.

You need to find a solution that will be really simple for the company's IT administrator to deploy a new website on the Hyper-V:

- No system installation or heavy configurations
- Only name, IP and site import configurations
- Minimal OS footprint and resource usage on the Hyper-V hosts

3.3 - Monitoring

Core infrastructure must include an easy monitoring solution. On the Hyper-V host, configure performance alerts on CPU and RAM usage. These alerts should send an email to the IT administrator. To facilitate error resolution in the infrastructure, centralize all logs on one server.

3.4 - Remote access administration

Administrator should be able to manage the infrastructure from a remote location if necessary. You will suggest 2 options:

- VPN server with RRAS
- Remote Desktop Service role

3.5 - Sites communication

As you know, *MedSearch* has two sites. Currently, they cannot have a redundant datacenter in Saint-Cénéri-le-Gerei.

Therefore, employees in Saint-Cénéri-le-Gerei must access the data and services available in the Caen datacenter. Data is extremely sensitive and must be accessed securely and without configuration on employee computers.

3.6 - Special request

A special request of the IT administrator will ask you some development. He wants to have an overview of all servers' health and events in a Web page:

- Map of all servers with icons indicators (events alert, ping response, etc.)
- Last 5 warning and critical events of each server
- For each warning, a button that will send an email to your company helpdesk with the content of the warning

4 - Delivery

Your rendering will be a ZIP containing your different configurations and deployments, a diagram of the infrastructure and a justification of your implementation choices.

You will show your work and give a live demonstration during the oral sessions. Your PowerPoint presentation will illustrate the technical aspects, you will also use a test station to show the functionality of your realizations.

5 - Grading scale

- High availability: 30 points

- IIS websites: 15 points
- Monitoring: 15 points
- Remote access administration: 12 points
- Sites communication: 13 points
- Special request: 15 points